CSE-476: Introduction to Natural Language Processing
Quiz 3

Multiple Choice Questions (Single correct)

Instructions: Select the correct option with a short explanation.

Q1. In a standard RNN, what is the primary purpose of the ``hidden state''?

A. To store the final output of the entire sequence.
B. To act as a memory that carries information from previous time steps
C. To calculate the loss function for the current word
D. To normalize the input embeddings

Q2. Why is the Softmax function typically used in the final layer of a Neural Language Model?

A. To ensure the output values sum to 1, representing a probability distribution
B. To prevent the gradients from exploding
C. To increase the speed of the training process
D. To allow the model to handle variable-length sequences

Q3. What is a key limitation of feed-forward neural networks for NLP tasks?

A. They cannot use embeddings
B. They assume fixed input size and struggle with temporal order
C. They cannot be trained with gradient descent
D. They do not use activation functions

Q4. What is the purpose of the ``Forward Pass'' in training?

A. To update weights based on the loss
B. To calculate the predicted output and compute the loss
C. To unroll the network into a sequence
D. To calculate the gradient for each parameter

Q5. What is true about the weights $U, V,$ and W ac ross different timestamps ina n RNN?

A. They are randomly re-initialized at every step
B. They are reused across timestamps
C. They increase in value as the sequence progresses
D. Only the output weight V is reused

Q6. Suppose all activation functions in a 3-layer neural network are linear. What is the effective model equivalent to?

A. A deeper network with more representational power
B. A single linear model
C. A recurrent neural network
D. A decision tree

Q7. When training an RNN for language modeling using teacher forcing, the model predicts:

A. The next word given the predicted history
B. The next word given the ground-truth history
C. The current word given itself
D. All words simultaneously

Q8. In an RNN, the hidden state at time t depends on:

A. Only the input at time t
B. Only the hidden state at time $t-1$
C. Both the input at time t and the hidden state at time $t-1$
D. All future inputs

Q9. Consider a neural network layer defined as: $h = f(Wx)$. The input vector x has shape $(n_0 \times 1)$, and the output vector h has shape $(n_1 \times 1)$. What must be the shape of W?

A. $(n_0 \times n_1)$
B. $(n_1 \times n_0)$
C. $(n_0 \times 1)$
D. $(1 \times n_1)$

Q10. What is the main role of backpropagation in neural network training?

A. To compute predictions during inference
B. To calculate gradients of the loss with respect to parameters
C. To initialize weights randomly
D. To normalize input data